Reference 03: Inverter & Charge Controller Costs

1. What is this? (Component & Checked Cost)

- **Component:** Fronius IG 40 Grid/Off-Grid Inverter ($4.4\ \mathrm{kW}$ DC / $3.5\ \mathrm{kW}$ AC) + MPPT Charge Controller & Protection
- **Checked Cost:** $\$1,200\ \mathrm{USD}$
- **Expected Lifetime:** $10–12\ \mathrm{years}$

2. The Proof (Screenshot & Commercial Quotes)

PV Installed Cost Benchmarks

Figure ES-1 compares our Q1 2023 MSP and MMP benchmarks for PV systems in the residential, community solar, and utility-scale sectors. The MMP benchmark is higher than the MSP benchmark for all sectors, because the MMP benchmark captures the inflationary market distortions that occurred in Q1 2023.

Our MMP benchmark for an 8-kW_{dc} residential PV system ($2.68 per watt direct current [W_{dc}]) is 15% higher than the MSP benchmark ($2.34/$W_{dc}$) and 15% lower than our MMP benchmark ($3.18/$W_{dc}$) from Q1 2022 in 2022 U.S. dollars (USD).

Our MMP benchmark for a 3-MW_{dc} fixed-tilt community solar system ($1.76/$W_{dc}$) is 18% higher than our MSP benchmark ($1.49/$W_{dc}$). Our Q1 2022 benchmark report has no community solar system for comparison.

Our MMP benchmark for a 100-MW_{dc} utility-scale system with one-axis tracking ($1.16/$W_{dc}$) is 19% higher than our MSP benchmark ($0.97/$W_{dc}$) and 8% higher than its counterpart ($1.07/$W_{dc}$) in Q1 2022 in 2022 USD. Compared with Q1 2022, higher inverter and EBOS costs plus new network upgrade costs more than offset lower module and SBOS costs in Q1 2023.

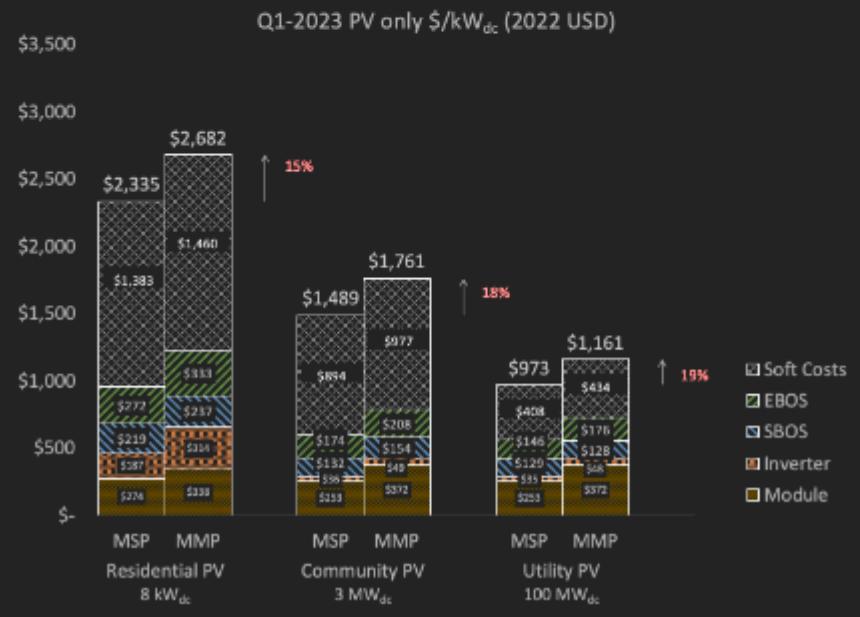


Figure ES-1. Q1 2023 U.S. PV cost benchmarks

Figure 3: NREL Q1 2023 U.S. Solar Photovoltaic System Cost Benchmark (Figure ES-1).

> **NREL Benchmark Data:** Inverter hardware represents $\$0.21–\$0.25/\mathrm{Wdc}$ in standalone commercial and residential solar systems. For our $4.41\ \mathrm{kWp}$ system:
>
> $$4,410\ \mathrm{W} \times \$0.25/\mathrm{W} \approx \$1,102\ \mathrm{USD}$$

> **Commercial Retail Quote:**
>
> - Supplier: SolarTopStore - Fronius IG Plus 3.0-1
> - Price: **£929.00 GBP** ($\approx \mathbf{\$1,200}\ \mathrm{USD}$ at $1.30\ \mathrm{GBP/USD}$).

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3. How was it calculated?

- **Base Inverter Cost:** Commercial quote of $\$1,100–\$1,200$ for Fronius IG units.
- **Integrated MPPT / Controller Overhead:** $\$100–\$200$.
- **Total Combined Allocation:** $\$1,200\ \mathrm{USD}$.
- **Comparison to Generative AI Guess:**
  - AI estimated $\$800$ for inverter + controller.
  - Generative AI underestimated the cost of European-standard transformer/MPPT equipment by $\sim 33\%$ ($+\$400$ adjustment required).

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4. Source Link

- **NREL Technical Report:** U.S. Solar Photovoltaic System and Energy Storage Cost Benchmarks: Q1 2023 (PDF)
- **Retail Hardware Quote:** SolarTopStore Fronius Inverter Catalog